

University of Dhaka
Faculty of Engineering & Technology
Department of Computer Science and Engineering

CSE 2216 - Application Development Lab (Batch 29)

Assignment 1 - Project Requirement Analysis

Team Name

Team Dev_Mavericks

Team Members

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Idea :

‘Murir Tin’ is a smart local bus app that uniquely combines color-coded routes, live GPS tracking, QR-based easy ticket booking, a complaint box, real-time crowd-powered updates, and emergency features — making daily transportation safer, smoother, and easier while putting control back in the passenger's hands.

Refer to Existing Applications:

(International)

1) BUSUP :

https://play.google.com/store/apps/details?id=com.busup.BusUp&pcampaignid=web_share

2) Busify :

https://play.google.com/store/apps/details?id=com.busify&pcampaignid=web_share

3) Moovit :

https://play.google.com/store/apps/details?id=com.tranzmate&pcampaignid=web_share

(Local)

4) Pathao :

https://play.google.com/store/apps/details?id=com.pathao.user&pcampaignid=web_share

App Screens:

<https://www.figma.com/design/mVImMeevgaXZZrFUYiG61D/Murir-Tin?t=f6OoY11xyBG7I9Mk-0>

Identify User Persona :

User Persona 1:

Name : Karim Bhuiyan

Age : 42

Gender: Male

Income Level : Low

Location: Savar

Profession : Garment Worker

Education Level : Low

Tech Proficiency : Basic (Can receive calls , use bkaash app . Not explored heavy high technological app)

Background :

Karim's workplace is in Mirpur. Every day, he uses public transport to reach his workplace. He cannot afford alternative rides like ride-sharing services or CNGs on a regular basis. Even when he uses public transport, there are times when the bus conductor charges more than the usual fare — sometimes an extra 10 to 30 taka. For Karim, that extra 10 to 30 taka is significant and hard to afford.

Need :

Karim requires a transportation system that offers fair and consistent pricing.

Feature:

Our app automatically calculates the fare between Karim's current location and destination . With just a tap of the "Pay Ticket" button and a quick PIN entry, Karim receives a QR code ticket, which eliminates the risk of fare overcharging and makes the payment process smooth and reliable.

User Persona 2 :

Name : Md. Ashiq Rahman

Age : 65

Gender : Male

Education Level : Medium

Tech Proficiency : Low (Can just give and receive calls)

Background :

Md. Ashiq Rahman is an old-aged person. He does not live in Dhaka. He often visits Dhaka to see her daughter and his grandchildren. As he does not have any close relatives in Dhaka, he travels alone. When he visits his daughter's house, he uses public transport. Since he is not a regular user of the public transportation system, he does not know which bus to go to. This lack of familiarity makes his travel experience stressful.

Need:

Md. Ashiq requires a simple way to navigate public transport routes, as well as assistance in finding the right bus and its route to ensure he reaches his destination safely and with minimal stress.

Feature:

Our app provides a user-friendly interface with color-coded routes that match the actual bus colors, making it easy for Md. Ashiq to identify the right bus at a glance. This color-coding system, combined with real-time updates, will help him navigate the transport system with ease and confidence, even if he's not a regular commuter.

User Persona 3

Name: Salhea Begum

Age : 32

Gender : Female

Education Level : High

Profession : Banker

Tech Proficiency : High

Background :

Salhea Begum is a banker. Her working place is quite far from her home. So taking CNG or ride-sharing apps is quite expensive for her. So she prefers to use public transport. But she tries to avoid overly crowded buses. It happened that she was waiting for the bus for a long time but when it arrived she saw the bus was already full. Besides that, sometimes she felt insecure while traveling on this public transport and went through harassment.

Need :

Salhea needs a safe, reliable public transport system that helps her avoid overcrowded buses and provides an easy way to report issues or request emergency help during her commute.

Feature:

Our app provides real-time crowd updates so Salhea can avoid overly crowded buses, and includes a complaint box and SOS emergency button to ensure safety and quick reporting of any harassment or issues during her travel.

User Persona 4:

Name : Rita Ghosh

Age :21

Gender : Female

Education Level : High

Tech Proficiency: High

Profession: University Student

Background:

Rita is a conscious citizen and a university student in Dhaka and has a tight class schedule. She often struggles to catch the right bus on time. It happened many times that she was waiting for the bus for a long time but the bus came late. Since time is important for her, waiting for the bus makes her day terrible. Besides that, as she is a

conscious citizen, the unfit bus and bus bus-driver 's ruthless driving makes her frustrated.

Need:

Rita needs a system that helps her track bus arrivals in real time and provides transparency about vehicle condition and driver behavior.

Feature:

Our app provides live gps tracking from where one can know where the bus's current location is. Moreover,if one faces any issue of reckless driving,unfit bus or anything else they can report their complain and also can view other people's reports.

Design Survey :

Question survey :

https://docs.google.com/forms/d/e/1FAIpQLSevBMU9I18kSFI-REIMBtltCGGMg29v_kowe02HrXHWfPOww/viewform?usp=header

Survey result :

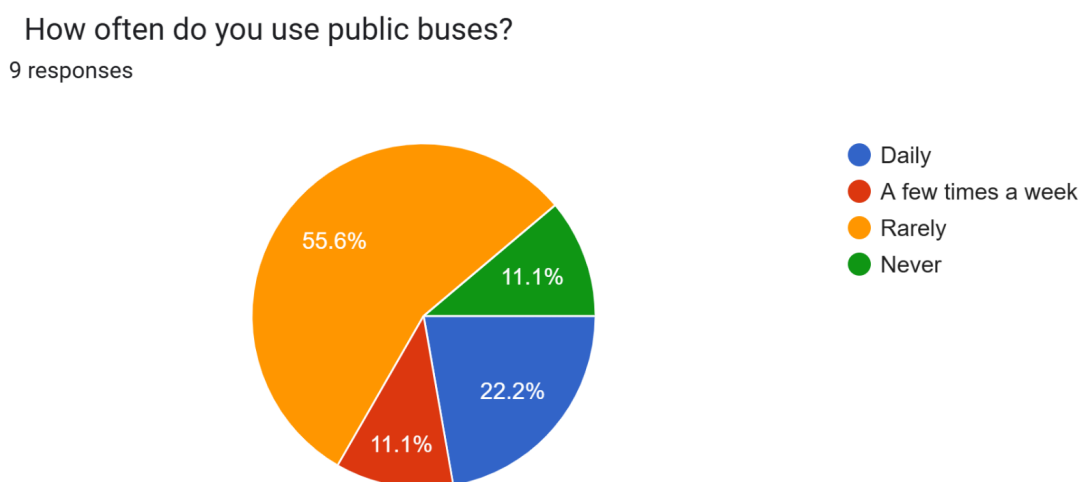


Fig-1 : Responses of use of Public Buses

What problems have you faced while using public transport? (Select all that apply)

9 responses

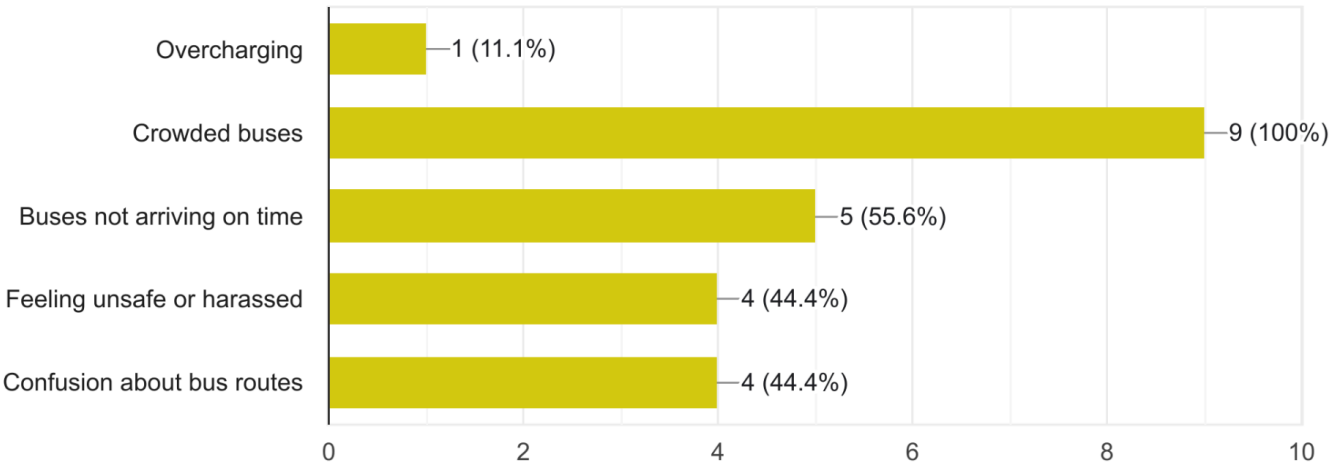


Fig-2 : Responses of facing problems of Public Transport

Do you currently use apps like bKash, Facebook, or Google Maps on your phone?

9 responses

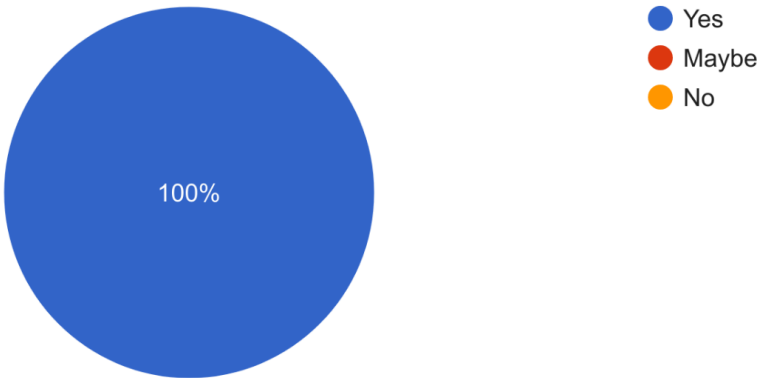


Fig-3 : Responses of Tech-Savvy People

Would you find it helpful if an app showed live bus locations and let you buy tickets with fair pricing?
9 responses

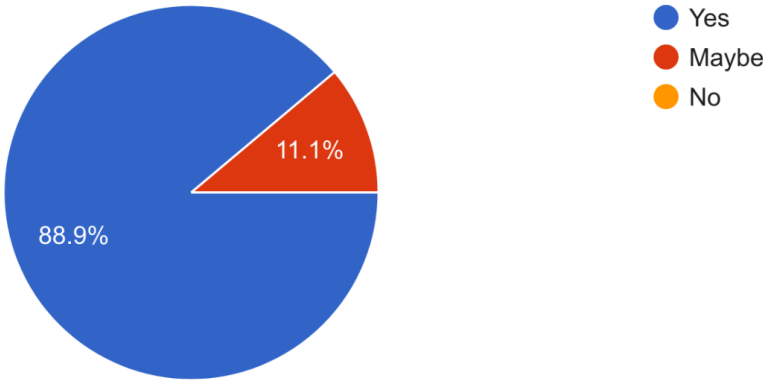


Fig-4 : Responses of interested people in app

How important is it for you to report problems like reckless driving or harassment while on a bus?
9 responses

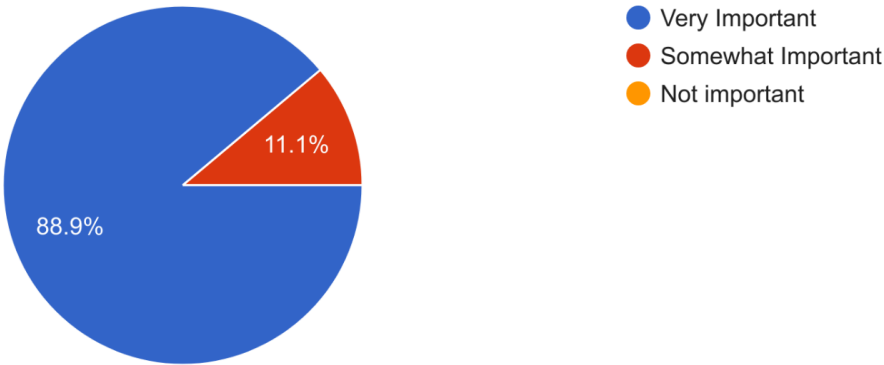


Fig-5: Responses of conscious people

Your Gender

6 responses

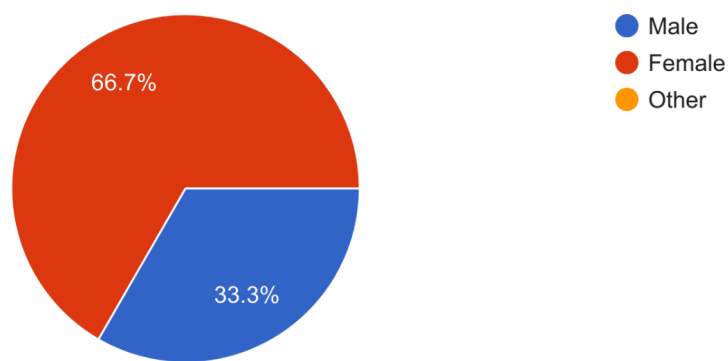


Fig -6 : Responses of Gender

Your age

9 responses

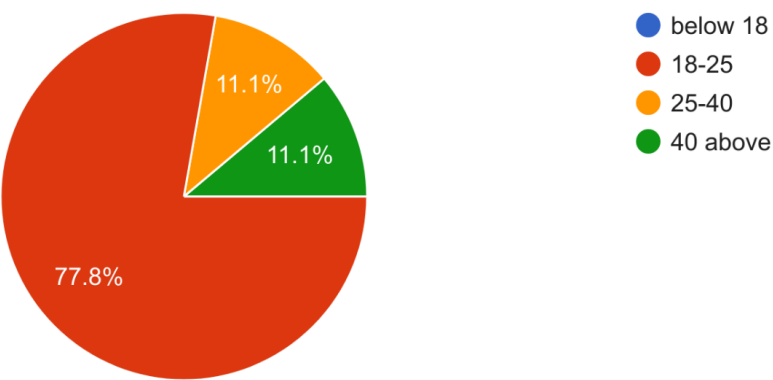


Fig-7 : Responses of Different age

Github Repository :

<https://github.com/srs4929/Mobile-App-lab.git>